

Aadi Soni

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EDUCATION

Centennial High School, Dallas, TX

August 2024 – Present

Aspiring Mechanical Engineer & Applied Mathematics Major

GPA 4.00/4.00

Relevant Coursework: AP Pre Calculus, AP Physics 1, AP Computer Science Principles, AP Statistics, AP Biology

EXPERIENCE

Sewell BMW Summer Engineering Specialist, Dallas, TX

May 2025 – September 2025 (30hrs/week)

Engineering Specialist & Performance Analysis

- Analyze automotive mechanical systems and performance data to optimize vehicle engineering processes and support diagnostic protocols in BMW service operations.

RESEARCH EXPERIENCE

University of Colorado Mathematical Machine Learning Research, Dallas, TX

May 2025 – Present (15hrs/week)

Applied ML for Mathematical Modeling & Prediction Systems

- Conducting machine learning research on mathematical modeling applications, developing predictive algorithms and computational methods for complex system analysis under faculty supervision.

University of Texas at Austin Applied Engineering Research, Dallas, TX

May 2025 – October 2025 (12hrs/week)

Clinical Application of Retinal Imaging Technology

- Conducted applied engineering research utilizing fundus photography for cardiovascular disease risk analysis, presented findings at MIT Undergraduate Research Technology Conference.

MORPH-RL - Morphing Wing Control System, Dallas, TX

September 2025 – December 2025 (10hrs/week)

Reinforcement Learning for Adaptive Aerodynamic Surfaces

- Designed an autonomous wing morphing control system using reinforcement learning to optimize aerodynamic performance across flight conditions, improving efficiency through adaptive geometry.

LEADERSHIP

C.R.O.S.S. (Collaboration in Research, Operations, Startups, and Science), Global

May 2025 – Present (8hrs/week)

Co-Founder & Executive President

- Founded global nonprofit with 100+ chapters worldwide, raising \$150K+ in funding to expand STEM education access and mentor 3000+ students in scientific research methods.

Centennial High School Science Olympiad & Bowl, Dallas, TX

October 2025 – Present (5hrs/week)

Build Events Captain

- Lead build event preparations and team training across multiple engineering competitions, developing comprehensive study materials and conducting tryouts for incoming members.

National Inventors Hall of Fame, Dallas, TX

April 2024 – Present (3hrs/week)

Leadership Intern

- Support STEM education by mentoring young learners in innovation and entrepreneurship while developing my communication skills through structured volunteer experience

SKILLS

Technical Skills/Software: Python (TensorFlow, scikit-learn, OpenCV, Matplotlib), R, MATLAB, SQL, GitHub, LaTeX

Professional Skills: Data Analysis, Critical Thinking, Quantitative Reasoning, Problem-Solving, Engineering Design

HONORS AND ACCOLADES

Centennial High School Salutatorian, Highest Honor – 2/474

2026

USA Physics Olympiad (USAPhO) Qualifier, ~Top 3% of ~8,000

2026

USA Computing Olympiad (USACO) Platinum Division, ~Top 2% of ~20,000

2026

MIT Undergraduate Research Technology Conference Presenter, Cardiovascular Disease Detection

2025

Texas State Science & Engineering Fair Finalist, Highest Honor – American Statistical Association, 1/~75

2024